Brevan Ellefsen

Mathematics Department
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Advisor: Nathan Dunfield

Education

UIUC — Champaign, Illinois
PhD Candidate — 2021–Present

Brigham Young University — Provo, Utah
B.S. in Mathematics, Major GPA: 4.00/4.00 — 2017–2020

Utah Valley University — Orem, Utah
A.S., GPA: 4.00/4.00 — 2015–2017

Research Experience

3-Manifold Theory — Champaign, Illinois
PhD Candidate advised by Nathan Dunfield — 2023-Present

- Proved a conjecture of Dunfield et al. regarding the precise asymptotic growth rate of essential surfaces in a Haken 3-manifold (paper to come)
- Extensive readings on 3-manifold and foliation theory

Knot Theory — Provo, Utah
Undergraduate research advised by Mark Hughes — 2019-2020

- Improved bounds on the petal numbers of torus knots and found presentations explicitly realizing these bounds
- Constructed a natural presentation for the fundamental group of knot complements from their petal diagrams
- Wrote extensive Python code for comparing knots by their invariants.

Complex Variables — Provo, Utah
Undergraduate research advised by David Cardon — 2017-2019

- Sought to characterize certain zero-preserving operators for sectors
- Conjectured a criterion for such operators via their actions on half planes
- Conjectured a method of constructing operators forcing roots into smaller sectors
- Found extensive numerical and visual evidence for both conjectures in Mathematica

Conference Talks

Introduction to Khovanov Homology — Provo, Utah
BYU Student Research Conference — 2019

- Earned prize for best presentation in group

Classification of Zero-Preserving Maps — Binghamton, New York
Binghamton University Graduate Conference in Algebra and Topology — 2019

Seminar Talks

UIUC Geometry and Topology Seminar

- Introduction to Categorical Dynamics
- Introduction to Haken Manifolds
- Virtually Haken Conjecture
- Counting Simple Closed Curves

Teaching

- TA for Calculus 1 (2x at UIUC)
- TA for Calculus 2 (4x at UIUC)
- TA for Linear Algebra (1x at UIUC)
- TA for Complex Analysis (1x at BYU)